

MF26 Formula List

List of Formulae and Statistical Tables · Singapore-Cambridge A-Level · used 2017 to 2024

H1 Mathematics (8865), H2 Mathematics (9758), H3 Mathematics and H2 Further Mathematics (9649)

Algebraic series

H2 Maths

Binomial expansions and Maclaurin series — Sequences & Series and the small- x approximations.

Binomial expansion, n a positive integer

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \binom{n}{3}a^{n-3}b^3 + \dots + b^n$$

where

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Maclaurin series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + \dots$$

Binomial series, any n

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots + \frac{n(n-1)\dots(n-r+1)}{r!}x^r + \dots$$

valid for $|x| < 1$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^r}{r!} + \dots$$

valid for all x

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^r x^{2r+1}}{(2r+1)!} + \dots$$

valid for all x

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^r x^{2r}}{(2r)!} + \dots$$

valid for all x

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{(-1)^{r+1} x^r}{r} + \dots$$

valid for $-1 < x \leq 1$

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AP: nth term and sum

$$u_n = a + (n - 1)d, \quad S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

GP: nth term and sum

$$u_n = ar^{n-1}, \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

GP: sum to infinity

$$S_\infty = \frac{a}{1 - r}$$

valid for $|r| < 1$

Sum and nth term

$$u_n = S_n - S_{n-1}$$

Small-angle approximations (x in radians)

$$\sin x \approx x, \quad \cos x \approx 1 - \frac{x^2}{2}, \quad \tan x \approx x$$

Expanding $(a + bx)^n$: take out a^n first

$$(a + bx)^n = a^n \left(1 + \frac{b}{a}x\right)^n$$

valid for $\left|\frac{b}{a}x\right| < 1$

Partial fractions

H2 Maths

Splitting a proper fraction before you integrate it or expand it as a series.

Distinct linear factors

$$\frac{px + q}{(ax + b)(cx + d)} = \frac{A}{ax + b} + \frac{B}{cx + d}$$

A repeated linear factor

$$\frac{px^2 + qx + r}{(ax + b)(cx + d)^2} = \frac{A}{ax + b} + \frac{B}{cx + d} + \frac{C}{(cx + d)^2}$$

A quadratic factor that does not factorise

$$\frac{px^2 + qx + r}{(ax + b)(x^2 + c^2)} = \frac{A}{ax + b} + \frac{Bx + C}{x^2 + c^2}$$

NOT ON MF26 — MEMORISEImproper fraction (top degree $\geq$ bottom degree): divide first

$$\frac{x^2 + 1}{(x - 1)(x + 2)} = 1 + \frac{-x + 3}{(x - 1)(x + 2)}$$

Trigonometry

H2 Maths

Compound and double angles, the factor formulae, and principal values.

MF27 keeps everything here except the four factor formulae.

$$\sin(A \pm B) \equiv \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) \equiv \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) \equiv \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A \equiv 2 \sin A \cos A$$

$$\begin{aligned}\cos 2A &\equiv \cos^2 A - \sin^2 A \\ &\equiv 2\cos^2 A - 1 \\ &\equiv 1 - 2\sin^2 A \\ \tan 2A &\equiv \frac{2\tan A}{1 - \tan^2 A}\end{aligned}$$

Factor formulae

$$\begin{aligned}\sin P + \sin Q &\equiv 2\sin \frac{1}{2}(P + Q) \cos \frac{1}{2}(P - Q) \\ \sin P - \sin Q &\equiv 2\cos \frac{1}{2}(P + Q) \sin \frac{1}{2}(P - Q) \\ \cos P + \cos Q &\equiv 2\cos \frac{1}{2}(P + Q) \cos \frac{1}{2}(P - Q) \\ \cos P - \cos Q &\equiv -2\sin \frac{1}{2}(P + Q) \sin \frac{1}{2}(P - Q)\end{aligned}$$

Principal values:

$$\begin{aligned}-\frac{1}{2}\pi &\leq \sin^{-1} x \leq \frac{1}{2}\pi \\ &\text{valid for } |x| \leq 1 \\ 0 &\leq \cos^{-1} x \leq \pi \\ &\text{valid for } |x| \leq 1 \\ -\frac{1}{2}\pi &< \tan^{-1} x < \frac{1}{2}\pi\end{aligned}$$

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Pythagorean identities

$$\sin^2 A + \cos^2 A \equiv 1, \quad 1 + \tan^2 A \equiv \sec^2 A, \quad 1 + \cot^2 A \equiv \operatorname{cosec}^2 A$$

R-formula

$$a \sin \theta + b \cos \theta \equiv R \sin(\theta + \alpha), \quad R = \sqrt{a^2 + b^2}, \quad \tan \alpha = \frac{b}{a}$$

$\cos 2A$ turned round (for integrating $\sin^2$ and $\cos^2$)

$$\sin^2 A \equiv \frac{1}{2}(1 - \cos 2A), \quad \cos^2 A \equiv \frac{1}{2}(1 + \cos 2A)$$

Derivatives

H2 Maths

Only the five awkward derivatives are given. Everything else in differentiation is yours to remember.

Unchanged in MF27.

$f(x)$	$f'(x)$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\sec x$	$\sec x \tan x$

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Basic derivatives

$$\frac{d}{dx} \sin x = \cos x, \quad \frac{d}{dx} \cos x = -\sin x, \quad \frac{d}{dx} \tan x = \sec^2 x, \quad \frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} \ln x = \frac{1}{x}, \quad \frac{d}{dx} a^x = a^x \ln a$$

Product and quotient rules

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}, \quad \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Chain rule and parametric form

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}, \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Integrals

H2 Maths

The eight standard forms. Constants of integration are left out and a is a positive constant.**Unchanged in MF27.**

$f(x)$	$\int f(x) dx$	Valid for
$\frac{1}{x^2 + a^2}$	$\frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$	
$\frac{1}{\sqrt{a^2 - x^2}}$	$\sin^{-1} \left(\frac{x}{a} \right)$	$ x < a$
$\frac{1}{x^2 - a^2}$	$\frac{1}{2a} \ln \left(\frac{x-a}{x+a} \right)$	$x > a$
$\frac{1}{a^2 - x^2}$	$\frac{1}{2a} \ln \left(\frac{a+x}{a-x} \right)$	$ x < a$
$\tan x$	$\ln(\sec x)$	$ x < \frac{1}{2}\pi$
$\cot x$	$\ln(\sin x)$	$0 < x < \pi$
$\operatorname{cosec} x$	$-\ln(\operatorname{cosec} x + \cot x)$	$0 < x < \pi$
$\sec x$	$\ln(\sec x + \tan x)$	$ x < \frac{1}{2}\pi$

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Basic integrals

$$\int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1), \quad \int \frac{1}{x} dx = \ln |x|, \quad \int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad \int \cos ax dx = \frac{1}{a} \sin ax, \quad \int \sec^2 ax dx = \frac{1}{a} \tan ax$$

Standard forms

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)|, \quad \int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1}$$

Integration by parts

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

Volume of revolution

$$V = \pi \int_a^b y^2 dx \quad (\text{about the } x\text{-axis}), \quad V = \pi \int_c^d x^2 dy \quad (\text{about the } y\text{-axis})$$

Vectors

H2 Maths

The ratio theorem and the cross product. Lines, planes, angles and distances are not given.

Unchanged in MF27.

The point dividing AB in the ratio $\lambda : \mu$

$$\frac{\mu \mathbf{a} + \lambda \mathbf{b}}{\lambda + \mu}$$

Vector (cross) product

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

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Scalar product

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Length of projection of a on b, and area of triangle

$$|\mathbf{a} \cdot \hat{\mathbf{b}}|, \quad \text{Area} = \frac{1}{2} |\mathbf{a} \times \mathbf{b}|$$

Line and plane

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}, \quad \mathbf{r} \cdot \mathbf{n} = D$$

Distance from point P to the plane $\mathbf{r} \cdot \mathbf{n} = D$

$$\frac{|\mathbf{p} \cdot \mathbf{n} - D|}{|\mathbf{n}|}$$

Angle between line and plane, and between two planes

$$\sin \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}| |\mathbf{n}|}, \quad \cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|}$$

Numerical methods

Further Maths

Further Maths: estimating an integral, a root, or a step of a differential equation. Not in H2 Maths.

Unchanged in MF27.

Trapezium rule, one strip

$$\int_a^b f(x) dx \approx \frac{1}{2}(b-a)[f(a) + f(b)]$$

Simpson's rule, two strips

$$\int_a^b f(x) dx \approx \frac{1}{6}(b-a) \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Newton-Raphson, x_1 a first approximation to a root of $f(x) = 0$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Euler method, step size h

$$y_2 = y_1 + hf(x_1, y_1)$$

Improved Euler method, step size h

$$u_2 = y_1 + hf(x_1, y_1), \quad y_2 = y_1 + \frac{h}{2}[f(x_1, y_1) + f(x_2, u_2)]$$

Standard distributions

H1 & H2 Maths

H1 and H2 Maths use the binomial row only. Poisson, geometric and exponential are Further Maths.

Unchanged in MF27.

DISCRETE

Distribution of X	$P(X = x)$	Mean	Variance
Binomial B(n, p)	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Poisson Po(λ)	$e^{-\lambda} \frac{\lambda^x}{x!}$	λ	λ
Geometric Geo(p)	$(1-p)^{x-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$

CONTINUOUS

Distribution of X	p.d.f.	Mean	Variance
Exponential	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

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Expectation and variance

$$E(X) = \sum x P(X = x), \quad \text{Var}(X) = E(X^2) - [E(X)]^2$$

Linear combinations (X, Y independent for the variance)

$$E(aX + b) = aE(X) + b, \quad \text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$$

Sample mean (central limit theorem, n large)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ approximately}$$

Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \quad P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Independent events, and counting

$$P(A \cap B) = P(A)P(B), \quad {}^n P_r = \frac{n!}{(n-r)!}, \quad {}^n C_r = \frac{n!}{r!(n-r)!}$$

Sampling and testing

H1 & H2 Maths

Estimating the population variance, from one sample or from two samples pooled.

MF27 keeps the one-sample formula and drops the pooled one.

Unbiased estimate of population variance

$$\begin{aligned} s^2 &= \frac{n}{n-1} \left(\frac{\sum (x - \bar{x})^2}{n} \right) \\ &= \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) \end{aligned}$$

Unbiased estimate of the common population variance from two samples

$$s^2 = \frac{\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}$$

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Unbiased estimate of the mean

$$\bar{x} = \frac{\sum x}{n}$$

z-test statistic for a mean

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Regression and correlation

H1 & H2 Maths

r and the y-on-x line. Your GC gives both; the formula is for questions that hand you summary totals.

Unchanged in MF27.

Product moment correlation coefficient

$$\begin{aligned} r &= \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \\ &= \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right) \left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}} \end{aligned}$$

Regression line of y on x

$$y - \bar{y} = b(x - \bar{x}), \quad b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

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Regression line of x on y (not given — swap the roles)

$$x - \bar{x} = d(y - \bar{y}), \quad d = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2}$$

Both lines pass through the mean point

$$(\bar{x}, \bar{y})$$

Statistical tables

H1 & H2 Maths

MF26 printed four tables: the normal distribution function $\Phi(z)$, critical values of t, critical values of χ^2 , and the Wilcoxon signed rank test.**MF27 keeps only the Wilcoxon table. Normal values now come from your graphing calculator.**

Old papers that say "use the tables in MF26" can be done on the graphing calculator.

 $\Phi(z)$: normalcdf. Its inverse: invNorm.The t and χ^2 tables served Further Maths tests.

Wilcoxon signed rank test

Further Maths

Further Maths non-parametric testing. Reject the null hypothesis when T is at most the value in the table.

Unchanged in MF27.

P = sum of the ranks of the positive differences.

Q = sum of the ranks of the negative differences.

T = the smaller of P and Q.

Each entry is the LARGEST T that still rejects the null hypothesis at that level.

A dash: no T can reject at that level for that n.

CRITICAL VALUES OF T

n	One-tail 0.05 · Two-tail 0.1	One-tail 0.025 · Two-tail 0.05	One-tail 0.01 · Two-tail 0.02	One-tail 0.005 · Two-tail 0.01
6	2	0	—	—
7	3	2	0	—
8	5	3	1	0
9	8	5	3	1
10	10	8	5	3
11	13	10	7	5
12	17	13	9	7
13	21	17	12	9
14	25	21	15	12
15	30	25	19	15
16	35	29	23	19
17	41	34	27	23
18	47	40	32	27
19	53	46	37	32
20	60	52	43	37

Typeset by Adrian's Math Tuition from the list SEAB publishes. The yellow boxes are formulas the list does NOT give. In the exam, the official booklet is the one handed out. Online: adrianmathtuition.com/formulas/mf26